

$$\begin{aligned} & \frac{g_1^2}{16\pi^2} + \frac{1}{16\pi^2} \\ & g_1^2 g_2^2 + 7 g_2^4 + \frac{1}{12} \sum_p g_1^4 L_{F2,0} [m_p^d] - \frac{1}{2} g_1^2 \overline{y_d^{pr}} y_d^{pr} L_{F2,0} [m_d^r] + \frac{1}{4} \sum_p g_1^4 L_{F2,0} [m_p^r] - \\ & \overline{y_e^{pr}} y_e^{pr} L_{F2,0} [m_e^r] + \frac{1}{8} (g_1^2 + g_2^2) (2 \overline{y_e^{pr}} y_e^{pr} + \sum_p (g_1^2 - g_2^2)) L_{F2,0} [m_1^p] + \\ & 6 \overline{y_d^{pr}} y_d^{pr} (g_1^2 - 3 g_2^2) + 6 \overline{y_u^{pr}} y_u^{pr} (g_1^2 + 3 g_2^2) + \sum_p (g_1^4 - 9 g_2^4) L_{F2,0} [m_p^q] + \\ & g_1^4 L_{F2,0} [m_p^r] - g_1^2 \overline{y_u^{pr}} y_u^{pr} L_{F2,0} [m_u^r] + \frac{1}{2} g_1^4 L_{F2,2,-2} [m_1, \tilde{\mu}] + \\ & m_1^2 L_{F2,2,-1} [m_1, \tilde{\mu}] + \frac{5}{2} g_2^4 L_{F2,2,-2} [m_2, \tilde{\mu}] + \frac{1}{2} g_2^4 m_2^2 L_{F2,2,-1} [m_2, \tilde{\mu}] + \\ & \tilde{\mu} g_2^4 (\overline{C_{\phi102}} + C_{\phi102}) L_{F2,2,0} [m_2, \tilde{\mu}] - \frac{1}{3} m_2 \tilde{\mu} g_2^4 (\overline{C_{\phi102}} + C_{\phi102}) L_{F3,2,-1} [m_2, \tilde{\mu}] + \\ & (-\overline{a_d^{pr}} a_d^{pr} + \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) L_{F2,1,0} [m_d^r, m_q^p] + \\ & (-C_{\phi2^2} \overline{a_d^{pr}} a_d^{pr} + C_{\phi12} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) L_{F2,2,0} [m_d^r, m_q^p] - \\ & y_d^{sr} \overline{y_u^{st}} y_u^{pt} L_{F1,1,0} [m_d^r, m_u^t] + \frac{1}{2} g_1^2 (-\overline{a_e^{pr}} a_e^{pr} + \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) L_{F2,1,0} [m_e^r, m_1^p] + \\ & (-C_{\phi2^2} \overline{a_e^{pr}} a_e^{pr} + C_{\phi12} \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) L_{F2,2,0} [m_e^r, m_1^p] + \\ & \tilde{\mu}^2 \overline{a_e^{pr}} a_e^{pr} - g_1^2 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} L_{F2,1,0} [m_1^p, m_e^r] + \frac{1}{4} (g_1^2 - g_2^2) (\overline{a_e^{pr}} a_e^{pr} + \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) \\ & (-1,1,1 [m_1^p, m_e^r] + \frac{1}{2} (\tilde{\mu} \overline{y_e^{pr}} (\overline{C_{\phi102}} a_e^{pr} - (g_1^2 + g_2^2) - 2 C_{\phi12} \tilde{\mu} g_1^2 y_e^{pr}) + \\ & \overline{a_e^{pr}} (2 C_{\phi2^2} g_2^2 a_e^{pr} + C_{\phi102} \tilde{\mu} y_e^{pr} (-g_1^2 + g_2^2))) L_{F3,1,0} [m_1^p, m_e^r] + \\ & (C_{\phi2^2} \overline{a_e^{pr}} a_e^{pr} - C_{\phi12} \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr}) L_{F3,2,-1} [m_1^p, m_e^r] + \\ & \overline{y_e^{pr}} (3 C_{\phi2^2} a_e^{pr} (g_1^2 - 3 g_2^2) + C_{\phi102} \tilde{\mu} y_e^{pr} (6 g_1^2 - 5 g_2^2)) + \\ & \overline{y_e^{pr}} (\overline{C_{\phi102}} a_e^{pr} (6 g_1^2 - 5 g_2^2) + 3 C_{\phi12} \tilde{\mu} y_e^{pr} (3 g_1^2 - g_2^2))) L_{F4,1,-1} [m_1^p, m_e^r] + \\ & \overline{y_e^{pr}} (\overline{C_{\phi102}} a_e^{pr} (-3 g_1^2 + 2 g_2^2) + 3 C_{\phi12} \tilde{\mu} y_e^{pr} (-g_1^2 + g_2^2)) + \\ & \overline{a_e^{pr}} (3 C_{\phi2^2} a_e^{pr} (-g_1^2 + g_2^2) + C_{\phi102} \tilde{\mu} y_e^{pr} (-3 g_1^2 + 2 g_2^2)) L_{F5,1,-2} [m_1^p, m_e^r] + \\ & \overline{a_d^{pr}} a_d^{pr} (-2 g_1^2 + 3 g_2^2) - g_1^2 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} L_{F2,1,0} [m_q^p, m_d^r] + \\ & (-2 - g_2^2) (\overline{a_d^{pr}} a_d^{pr} + \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) L_{F3,1,-1} [m_q^p, m_d^r] + \\ & \overline{y_d^{pr}} (3 \overline{C_{\phi102}} a_d^{pr} (-g_1^2 + g_2^2) - 2 C_{\phi12} \tilde{\mu} g_1^2 y_d^{pr}) + \\ & \overline{a_d^{pr}} (2 C_{\phi2^2} a_d^{pr} (-2 g_1^2 + 3 g_2^2) + 3 C_{\phi102} \tilde{\mu} y_d^{pr} (-g_1^2 + g_2^2)) L_{F3,1,0} [m_q^p, m_d^r] + \\ & (C_{\phi2^2} \overline{a_d^{pr}} a_d^{pr} - C_{\phi12} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr}) L_{F3,2,-1} [m_q^p, m_d^r] + \\ & \overline{y_d^{pr}} (C_{\phi2^2} a_d^{pr} (7 g_1^2 - 9 g_2^2) + C_{\phi102} \tilde{\mu} y_d^{pr} (6 g_1^2 - 5 g_2^2)) + \\ & \overline{y_d^{pr}} (\overline{C_{\phi102}} a_d^{pr} (6 g_1^2 - 5 g_2^2) + C_{\phi12} \tilde{\mu} y_d^{pr} (5 g_1^2 - 3 g_2^2))) L_{F4,1,-1} [m_q^p, m_d^r] + \\ & \overline{y_d^{pr}} (\overline{C_{\phi102}} a_d^{pr} (-3 g_1^2 + 2 g_2^2) + 3 C_{\phi12} \tilde{\mu} y_d^{pr} (-g_1^2 + g_2^2)) + \\ & \overline{a_d^{pr}} (3 C_{\phi2^2} a_d^{pr} (-g_1^2 + g_2^2) + C_{\phi102} \tilde{\mu} y_d^{pr} (-3 g_1^2 + 2 g_2^2)) L_{F5,1,-2} [m_q^p, m_d^r] - \\ & y_d^{sr} \overline{y_u^{st}} y_u^{pt} L_{F1,1,0} [m_q^p, m_q^s] + \frac{1}{4} (g_1^2 + 3 g_2^2) (\overline{a_u^{pr}} a_u^{pr} - \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr}) \\ & (-1,0 [m_q^p, m_u^r] + \frac{1}{4} (\overline{a_u^{pr}} (C_{\phi12} a_u^{pr} (g_1^2 + 3 g_2^2) + 2 C_{\phi102} \tilde{\mu} g_2^2 y_u^{pr}) + \\ & \overline{y_u^{pr}} (2 \overline{C_{\phi102}} g_2^2 a_u^{pr} - C_{\phi2^2} \tilde{\mu} y_u^{pr} (g_1^2 + 3 g_2^2))) L_{F2,2,0} [m_q^p, m_u^r] - \\ & \tilde{\mu}^2 (\overline{C_{\phi102}} \overline{y_u^{pr}} a_u^{pr} + C_{\phi102} y_u^{pr} \overline{a_u^{pr}}) L_{F3,2,-1} [m_q^p, m_u^r] + \\ & \overline{y_u^{pr}} a_u^{pr} (-7 g_1^2 + 3 g_2^2) + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr} (g_1^2 + 3 g_2^2) L_{F2,1,0} [m_u^r, m_q^p] + \\ & (-2 - g_2^2) (\overline{a_u^{pr}} a_u^{pr} + \tilde{\mu}^2 \overline{y_u^{pr}} y_u^{pr}) L_{F3,1,-1} [m_u^r, m_q^p] + \\ & \overline{y_u^{pr}} (C_{\phi12} a_u^{pr} (-7 g_1^2 + 3 g_2^2) + C_{\phi102} \tilde{\mu} y_u^{pr} (-3 g_1^2 + 2 g_2^2)) + \\ & \overline{y_u^{pr}} (\overline{C_{\phi102}} a_u^{pr} (-3 g_1^2 + 2 g_2^2) + C_{\phi2^2} \tilde{\mu} y_u^{pr} (g_1^2 + 3 g_2^2)) L_{F3,1,0} [m_u^r, m_q^p] + \\ & \overline{a_u^{pr}} (C_{\phi12} a_u^{pr} (g_1^2 + 3 g_2^2) + 2 C_{\phi102} \tilde{\mu} g_2^2 y_u^{pr}) + \\ & \overline{y_u^{pr}} (-2 \overline{C_{\phi102}} g_2^2 a_u^{pr} + C_{\phi2^2} \tilde{\mu} y_u^{pr} (g_1^2 + 3 g_2^2)) L_{F3,2,-1} [m_u^r, m_q^p] + \\ & \overline{y_u^{pr}} (\overline{C_{\phi102}} a_u^{pr} (3 g_1^2 - 2 g_2^2) + C_{\phi2^2} \tilde{\mu} y_u^{pr} (g_1^2 - 3 g_2^2)) + \\ & \overline{a_u^{pr}} (C_{\phi12} a_u^{pr} (5 g_1^2 - 3 g_2^2) + C_{\phi102} \tilde{\mu} y_u^{pr} (3 g_1^2 - 2 g_2^2)) L_{F4,1,-1} [m_u^r, m_q^p] + \\ & \overline{y_u^{pr}} (\overline{C_{\phi102}} a_u^{pr} (-3 g_1^2 + 2 g_2^2) + 3 C_{\phi2^2} \tilde{\mu} y_u^{pr} (-g_1^2 + g_2^2)) + \\ & \overline{a_u^{pr}} (3 C_{\phi12} a_u^{pr} (-g_1^2 + g_2^2) + C_{\phi102} \tilde{\mu} y_u^{pr} (-3 g_1^2 + 2 g_2^2)) L_{F5,1,-2} [m_u^r, m_q^p] - \\ & y_d^{sr} \overline{y_u^{st}} y_u^{pt} (C_{\phi12} + C_{\phi2^2}) L_{F2,1,0} [m_u^t, m_d^r] + 3 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} \\ & (-2 + C_{\phi2^2}) L_{F3,1,-1} [m_u^t, m_d^r] + \frac{3}{4} (g_1^4 - g_1^2 g_2^2) L_{F2,1,-1} [\tilde{\mu}, m_1] + \\ & g_1^4 + g_1^2 g_2^2 L_{F3,1,-2} [\tilde{\mu}, m_1] + g_1^2 (C_{\phi12} + C_{\phi2^2}) (g_1^2 - g_2^2) L_{F3,1,-1} [\tilde{\mu}, m_1] - \\ & \tilde{\mu} g_1^2 (\overline{C_{\phi102}} + C_{\phi102}) (g_1^2 - g_2^2) L$$